Experiment No:20
Experiment to understand K-means clustering algorithm for a given data set.

AIM:
To understand K-means clustering algorithm for a given data set.

DESCRIPTION:
Understand the K-means clustering algorithm for the dataset given.

CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline

df=pd.read_csv('/content/Mall_Customers.csv')

df.info()
df.head()

sns.pairplot(df)

features=df.iloc[:,[3,4]].values

from sklearn.cluster import KMeans

model=KMeans(n_clusters=5,random_state=42,n_init=10)

model.fit(features)

Final=df.iloc[:,[3,4]].copy()
Final['label']=model.predict(features)
```

```
Final.head()

sns.set_style("whitegrid")

g=sns.FacetGrid(Final,hue="label",height=8)
g.map(plt.scatter,"Annual Income (k$)","Spending Score (1-100)")
g.add_legend()

plt.show()

features_el=df.iloc[:,[2,3,4]].values

wcss=[]

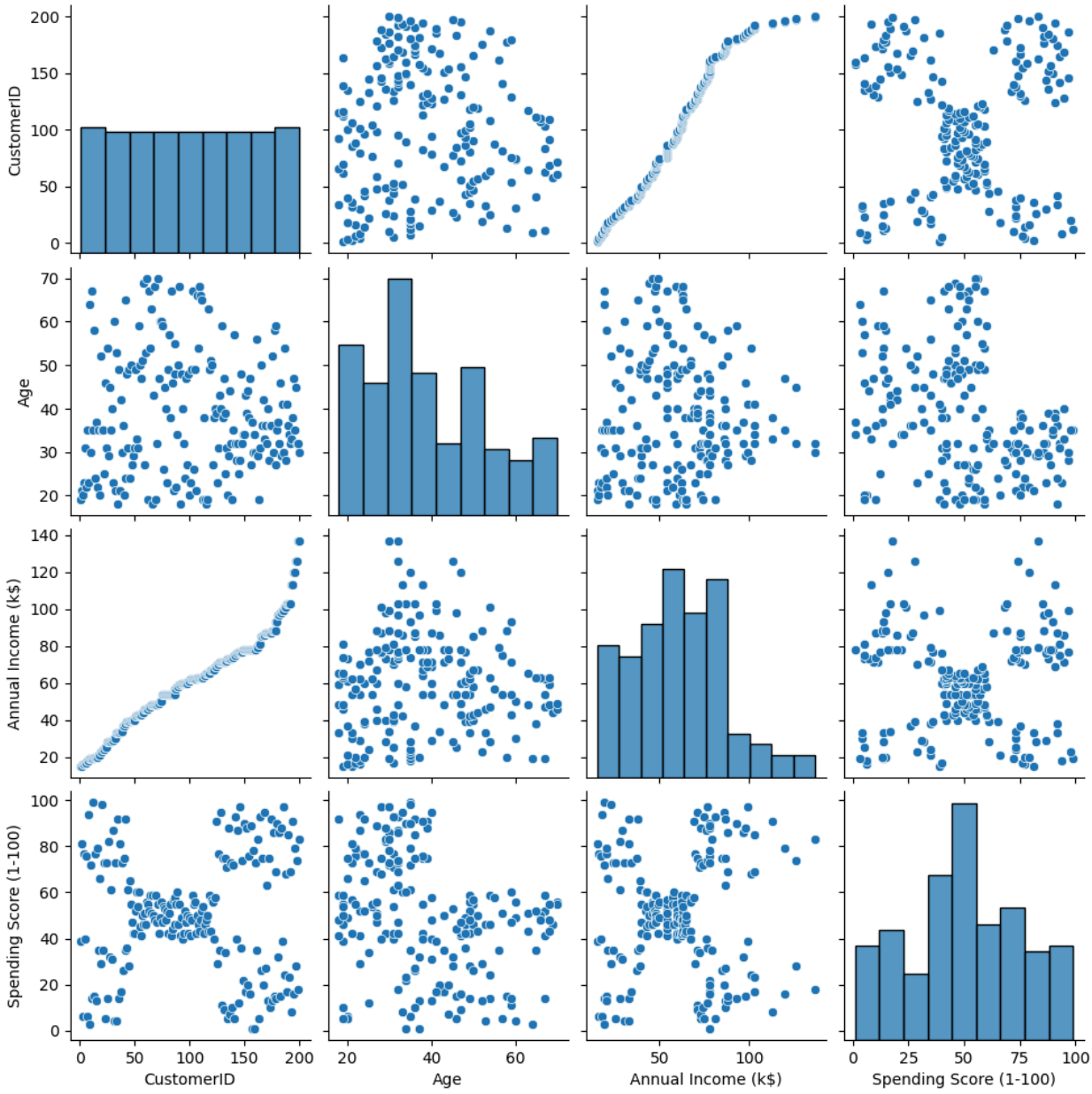
for i in range(1,10):
    model=KMeans(n_clusters=i,random_state=42,n_init=10)
    model.fit(features_el)
    wcss.append(model.inertia_)

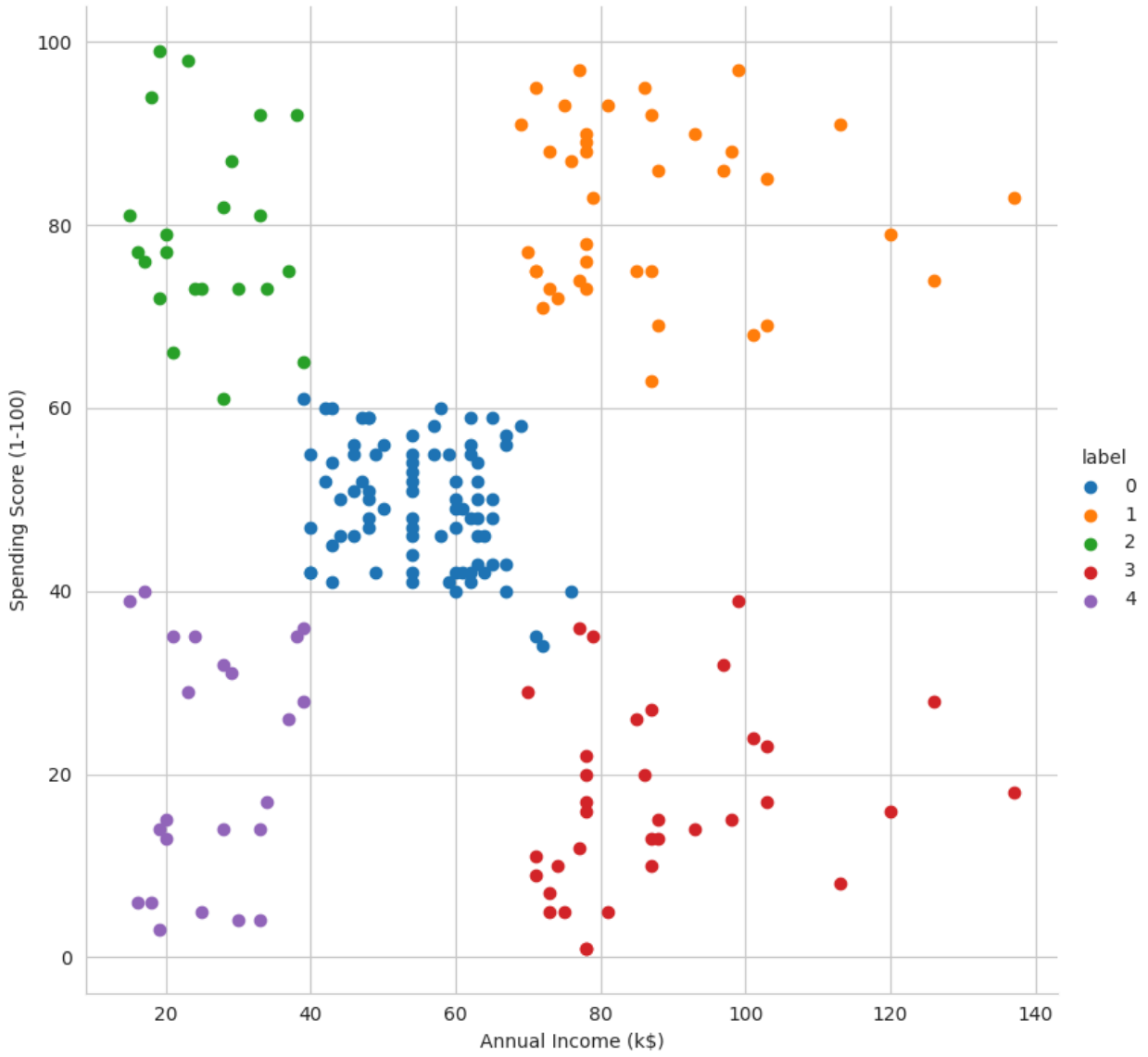
plt.plot(range(1,10),wcss,marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```

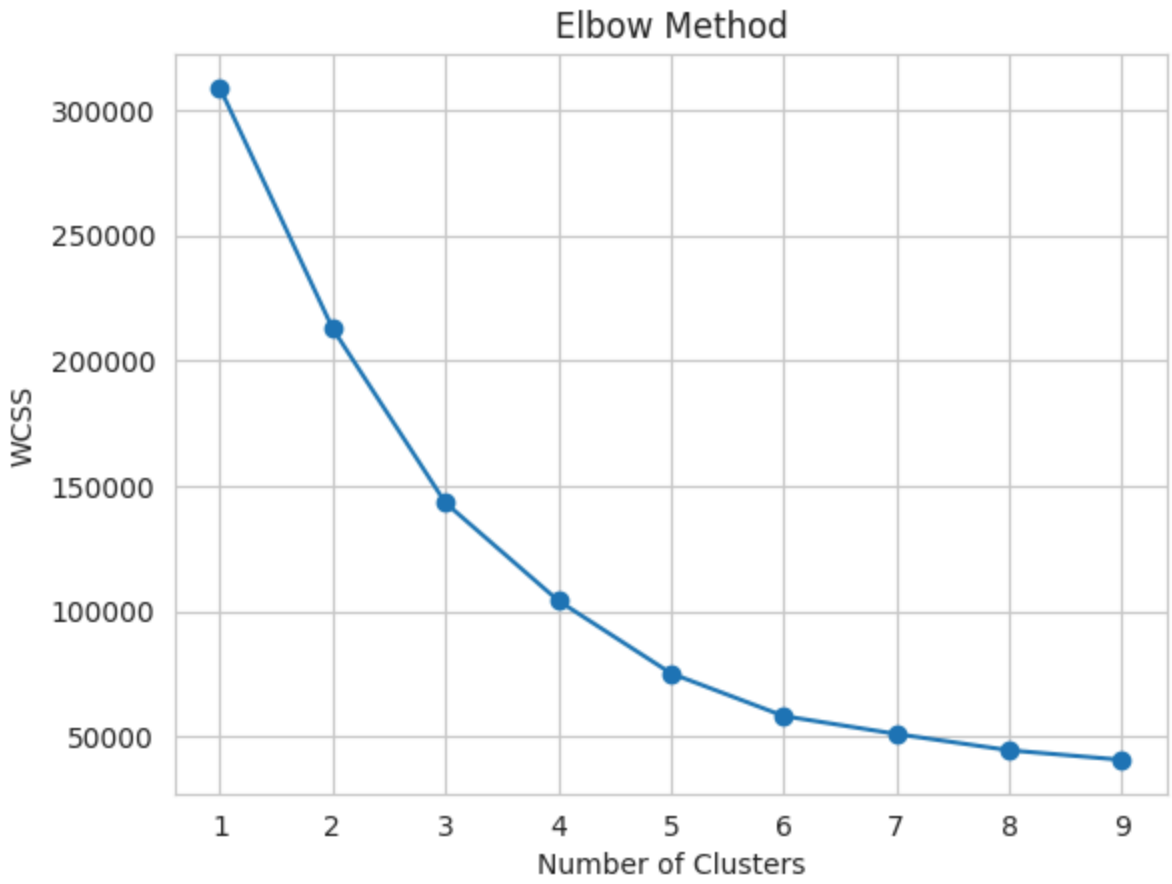
OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
```

```
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```




Spending Score (1-100)
Annual Income (k$)
label
0
1
2
3
4
0
20
40
60
80
100
20
40
60
80
100
120
140



RESULT:

Thus, the K-means clustering algorithm was successfully implemented on the Mall Customers dataset. The customers were grouped into five clusters based on their annual income and spending scores. The clusters were visualized using a scatter plot, and the optimal number of clusters was analyzed using the Elbow Method with the Within-Cluster Sum of Squares (WCSS) graph.